

HPML FINAL PROJECT · SPRING 2026

SmartGridBench

MCP-based profiling of tool-using LLM agents for Smart Grid transformer maintenance — measuring how transport, orchestration, and profiling change a fixed-8B agent.

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LINKS

GitHub: github.com/HPML6998-S26-Team13/hpml-assetopsbench-smart-grid-mcp

Tracking: wandb.ai/assetopsbench-smartgrid · IBM AssetOpsBench PR #287, #292

MENTOR

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Motivation & *Problem*

WHY THIS MATTERS

Transformer maintenance is a real industrial setting where wrong calls hit safety, cost, and outages.

A believable agent must retrieve telemetry, inspect DGA, run a forecast, then write a work order — four tools on one asset.

A fluent final answer is not enough; an unsupported recommendation can sound right and still be unsafe.

HPML pressure: tool-using inference adds cold starts, transport overhead, and replanning loops a naive baseline can't track.

PROBLEM STATEMENT

Profile a fixed **Llama-3.1-8B-Instruct** agent on a multi-tool Smart Grid workload, and measure how **transport** (direct → MCP → optimized MCP) and **orchestration** (Agent-as-Tool → Plan-Execute → Verified PE ± Self-Ask) change latency, GPU usage, and judged quality.

GOAL

A **measured tradeoff at iso-quality** on a single A100 — not a single speedup number. Transport, orchestration, and profiling must be reported separately so claims stay honest.

Technical Approach — Model & *Data*

MODEL

Llama-3.1-8B-Instruct

Self-hosted via **vLLM** on A100 — the core matrix.

JUDGE LLM

Llama-4-Maverick-17B on WatsonX · 6-criterion rubric · threshold 0.6

SCALE SANITY (APPENDIX)

Llama-3.3-70B hosted on WatsonX — never mixed into the 8B core.

DATA

Smart Grid transformer workload

License-clean: telemetry, DGA records, forecast tasks, WO fixtures.

CORPUS

61 canonical scenarios + **5** negative validation fixtures

EVIDENCE FLOOR

31-scenario post-PR175 set for most judged result tables — captions name which floor each row uses.

HARDWARE

A100-SXM4-40GB

Insomnia + GCP capture paths.

SOFTWARE

vLLM · PyTorch · MCP server (studio)

TRACKING

W&B project assetopsbench-smartgrid

UPSTREAM

IBM AssetOpsBench **PR #287** · review ready

IBM AssetOpsBench **PR #292** · review ready

Every cell holds the **model, judge, and data fixed** — the only things changing are transport, orchestration, and profiling coverage.

Technical Approach — *Optimizations* Applied

1. MCP transport standardization

Replace direct Python tool calls with a stdio MCP server — same tools, different protocol — to measure protocol overhead in isolation. A → B

2. Persistent + cached MCP transport

Reuse a long-lived MCP session across calls; cache repeated tool queries. Targets median latency without changing the agent. B → C / D

3. Verified Plan-Execute + Self-Ask

Add an explicit verification step plus Self-Ask clarification before the final answer — cuts unsupported answers without changing tools. B → Z → ZS

4. Full-stack corner: ZSD & D

Combine the persistent transport with Verified PE + Self-Ask to fill the corner of the matrix and check whether transport & orchestration gains compose.

EXPERIMENT MATRIX · CELLS RUN

	DIRECT	MCP	MCP +TP	+CACHE (D)
AGENT - AS - TOOL	A · AT-I	B · AT-M	C · AT-TP	D · AT-TPQ
PLAN - EXECUTE	—	Y · PE-M	YS-TP	—
VERIFIED PE	—	Z · V-M	ZS-TP	ZSD · V-S-TPQ
+ SELF - ASK	—	YS / ZS	↑	↑

B is the anchor — serves as both the MCP transport baseline and the Agent-as-Tool orchestration baseline.

We **filled the corners** with D (full transport stack) and ZSD (full orchestration + transport stack).

Every cell emits the same artifact contract.

Profiling *Methodology*

TOOLS

- Weights & Biases — run-level tracking, 63 / 104 rows linked.
- nvidia-smi — GPU util, VRAM, power on the four spot-check runs.
- PyTorch / vLLM traces — kernel-level timing where hooks fire.
- Llama-4-Maverick-17B judge — 6-criterion rubric, threshold 0.6.

METRICS CAPTURED

- Latency — p50, p95 per cell · per-trial `latencies.jsonl`.
- GPU utilization & peak VRAM — A100 telemetry.
- Per-stage time — tool-call → LLM forward → final-answer compose.
- Task quality — judge mean & pass-rate on the 31-scenario floor.

STABILITY GUARDRAILS

Same artifact contract

Every cell writes `summary.json`, `latencies.jsonl` , raw trial JSON. No cell is special.

Judged ≠ profiled

Profiler rows support observability claims; quality claims require judge rows under the rubric.

Floor named on every caption

31-scenario judged · 61-scenario corpus · 4 spot-check runs — never collapsed into one number.

Results — *Headline* Numbers

TRANSPORT · C VS B

–43%

p50 latency on the matched final-six A100 cohort, 13.09s → 7.40s.

ORCHESTRATION · ZS

0.833

judge mean & 5 / 6 pass on the 31-scenario floor — Verified PE + Self-Ask.

PROFILING INVENTORY

104

run rows · 63 W&B-linked · 13 profiler-linked · 6 with summary stats.

Three findings on **different evidence streams** — latency, judged quality, observability. We learned an HPML systems lesson along the way: they do not collapse into one speedup number.

All measurements averaged over 6 trials per cell

A100-SXM4-40GB

vLLM-served Llama-3.1-8B-Instruct

Error bars in backup

Results — *Before* vs. After

CELL	MEANING	P50	P95	JUDGE PASS
A · AT-I	Direct tools	12.15s	17.29s	1 / 6
B · AT-M	MCP baseline · anchor	13.09s	16.27s	2 / 6
C · AT-TP	Optimized MCP	7.40s	47.93s	0 / 6

P50 MEDIAN

Warm-path win. Persistent transport reuses the MCP session across tools.

P95 TAIL

Cold-start cost moves to the tail. First-call init dominates outliers.

Takeaway. Optimized MCP cuts p50 by 43% but expands the cold-tail p95 by ~3× and does *not* improve judged quality. Median speedup ≠ quality lift.

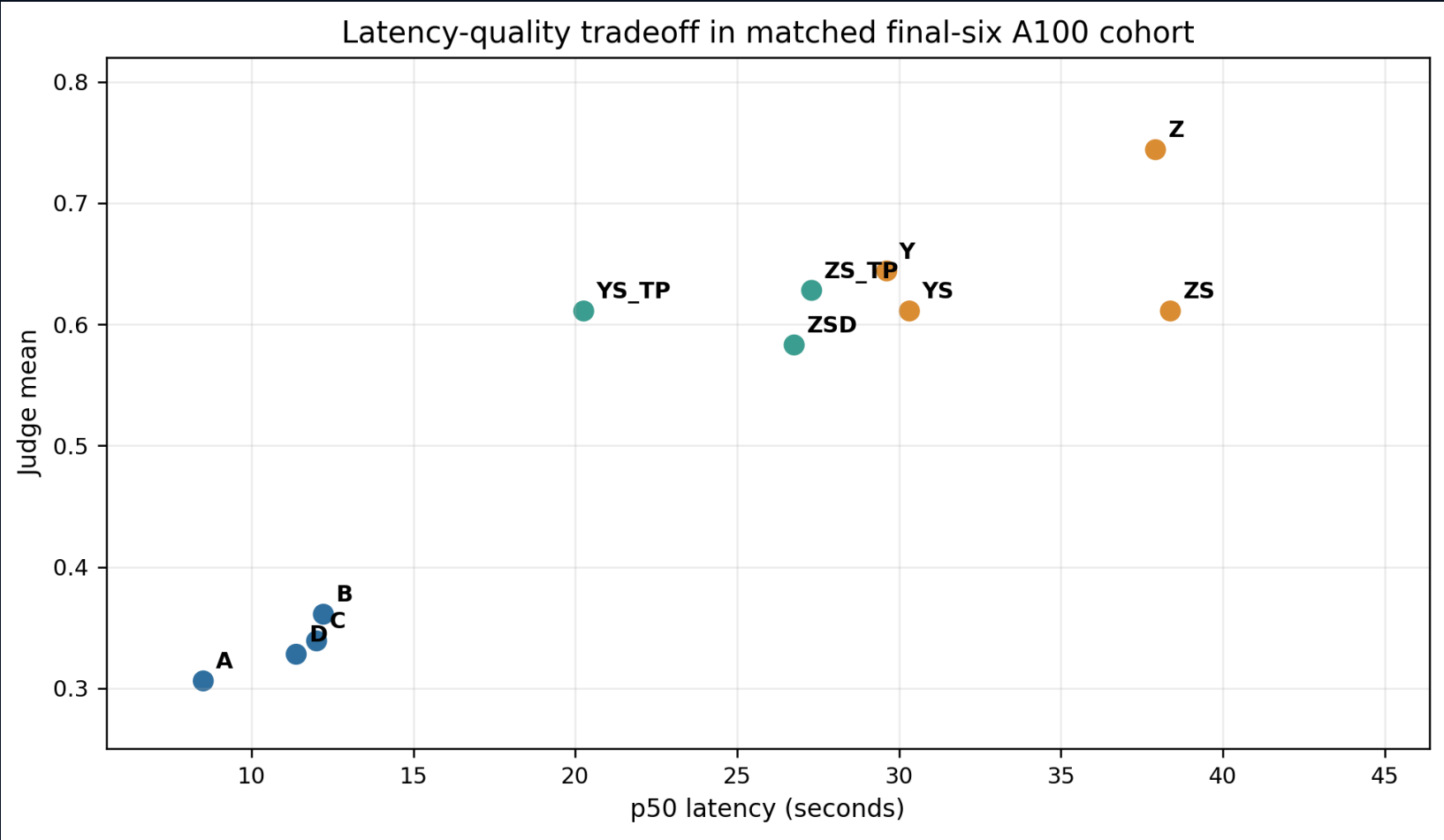


FIG · matched final-six A100 cohort · p50 latency × judge mean

B is the anchor. C improves p50 but lands in the same low-quality cluster – moving along latency, not quality.

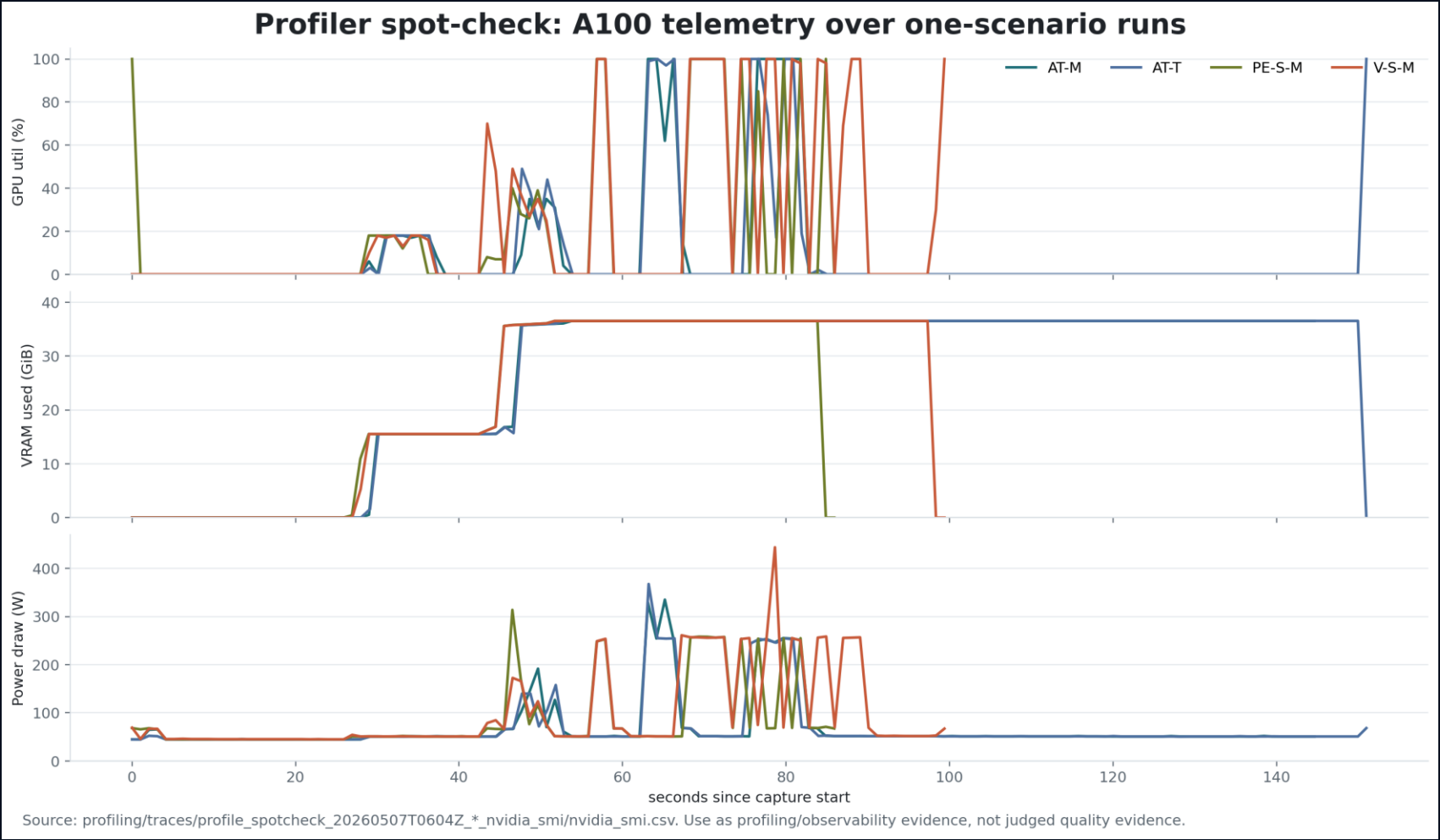
Profiler *Evidence*

SPOT-CHECK CELL	W&B	NVIDIA-SMI	TORCH TRACE
AT-M	✓	✓	✓
AT-T	✓	✓	✓
PE-S-M	✓	✓	X gap
V-S-M	✓	✓	X gap

FINDINGS

- Idle gap during MCP cold start that shrinks on AT-T.
- vLLM forward kernels dominate inside each tool turn.
- PE / V torch hooks did not fire — coverage gap, not failure.

Gap analysis. AT-M and AT-T emit torch traces; PE-S-M and V-S-M do not. Profiler-hook coverage gap, not a failed run — and a real next-step item.



A100 telemetry · GPU util / VRAM / power · profile_spotcheck_20260507T0604Z

Demo / Visualization

● ● ● scenario SGT-009 · cell ZS · trial 4 / 6 · judge=Maverick-17B

```
$ $ ./scripts/demo_static_playback.py --case all-domain --mode brief

[plan] load_telemetry(asset="SGT-009")
[plan] read_dga(asset="SGT-009")
[plan] forecast_load(asset="SGT-009", horizon="7d")
[plan] write_work_order(...)
tool   load_telemetry  → 17,280 samples  (2.1s)
tool   read_dga        → H2=412 ppm, C2H2=18 ppm  [elevated]
tool   forecast_load   → p95 1.18× nameplate [overload risk]
[verify] evidence  ✓  DGA cited · telemetry cited · forecast cited
[verify] recommend ✓  matches asset class · grounded

answer  Schedule oil sampling within 7 days. Reduce peak
        loading by 15%. DGA ratios indicate partial-
        discharge progression; forecast shows recurring
        overload window Wed-Fri.

judge   task_completion: True
judge   data_retrieval_accuracy: True
judge   generalized_result_verification: True
judge   agent_sequence_correct: True
judge   clarity_and_justification: True
judge   hallucinations: False
judge   pass
```

NOTE: SIMULATION

Plan-Execute issues a 4-tool plan on one transformer asset.

The verifier checks the answer cites every tool's evidence before finalizing.

Judge scores six dimensions; mean 0.833 on the 31-scenario floor.

Caveat. The 8B run demonstrates full cross-domain tool use, but overstates the diagnosis. The grounded answer is probable low-temperature overheating, mixed DGA evidence, RUL 547 -> 517 days, and a high-priority inspection work order. A 70B run emits this conclusion correctly, consistent with production settings where larger/frontier models are preferred for high-stakes operational decisions.

data/scenarios/SGT-009.json → benchmarks/cell_ZS/raw/<run-id>/ → results/metrics/zs.csv

Lessons *Learned*

WHAT WORKED

MCP standardization is measurable — separates transport from orchestration cleanly without changing the agent.

Verified PE + Self-Ask reaches the strongest judged row (ZS, mean 0.833) — verification & clarification matter more than plan structure.

One artifact contract across all cells kept comparisons honest under deadline pressure.

WHAT DIDN'T WORK

Persistent MCP regressed. p50 dropped 43% but cold-tail p95 expanded ~3× and judge pass went to 0 / 6.

Vanilla Plan-Execute was weak against the AaT anchor — judge mean 0.111 vs 0.278.

Hosted 70B (V-M-70) is not a free win. Matched 8B judge mean but p95 ≈ 271s vs ≈ 58s — scale isn't auto-win on this workload.

Most failures we audited are **evidence failures, not plumbing failures** — a run can finish and still miss the task. That is why judge rows and profiler rows must be reported separately.

Next *Steps*

1. Refresh judged matrix on the full 61-scenario corpus.

Close the gap between the 61-scenario corpus and the 31-scenario judged floor so corpus truth and result truth match.

2. Close the PE / Verified-PE torch-trace hook gap.

Profiler coverage should be even across orchestration modes — currently only AT-M / AT-T emit torch traces.

3. Shepard IBM AssetOpsBench PR #287 to merge.

PR is ready for maintainer review: non-draft, DCO green, mergeable; waiting on IBM review.

4. Upstream the 25 post-submission scenarios. Opened IBM AssetOpsBench PR #292.

PR #199 stretch corpus — 25 additional canonical scenarios drafted after submission. Land them on top of the 61-scenario base so the public benchmark covers more failure modes.

? Open question.

Hosted 70B (V-M-70) matched 8B judge mean but at p95 ~271s vs ~58s. Can a smaller verifier reach ZS quality at AT-M latency? That is the next inference-systems study.

Teamwork & *Contributions*

Akshat Bhandari

Scenarios, judging, and failure audit

- Authored the 61-scenario canonical corpus + 5 negative validation fixtures.
- Built the judge/eval harness around Llama-4-Maverick (6-criterion, 0.6 threshold).
- Generated-scenario validation and failure-taxonomy audit.
- Drafted slides 9–10 (demo path, lessons learned).

Aaron Fan

Infrastructure and profiling

- Insomnia + GCP + vLLM A100 serving stack and runbooks.
- Capture scripts, profiling wrappers, W&B schema integration.
- A/B transport evidence (AT-M vs AT-T) and the 4 spot-check profile rows.
- Drafted slides 5–8 (profiling, headline, before/after, profiler evidence).

Tanisha Rathod

Data pipeline and MCP servers

- End-to-end data pipeline: telemetry · DGA · forecasts · work-order fixtures.
- Implementation of all 4 MCP servers and hardening across 4 tool domains.
- Standards artifacts (artifact contract, schema docs) and methodology support.
- Drafted slides 3–4 (model & data, optimizations applied).

Wei Alexander Xin

Coordination and analysis

- Project coordination, experiment-matrix design, deadline management.
- Runner / W&B schema, unified artifact contract.
- Orchestration & mitigation analysis (Verified PE + Self-Ask story).
- Drafted slides 1–2, 11–14; owned upstream IBM AssetOpsBench PR #287, #292, and final deliverables.

Collaboration: weekly syncs, shared WhatsApp channel, code review on every PR. Currently awaiting PR #287 in IBM AssetOpsBench.

Thank *you.*

Questions?

REPO

[github.com/HPML6998-S26-Team13/
hpml-assetopsbench-smart-grid-mcp](https://github.com/HPML6998-S26-Team13/hpml-assetopsbench-smart-grid-mcp)

TRACKING

wandb.ai/assetopsbench-smartgrid

UPSTREAM

github.com/IBM/Asset0psBench/pull/287
github.com/IBM/Asset0psBench/pull/292

Backup — *Experimental* Setup

SETTING	CORE MATRIX	SCALE-SANITY (APPENDIX)
Model	Llama-3.1-8B-Instruct	Llama-3.3-70B
Serving	vLLM, self-hosted	WatsonX hosted endpoint
Hardware	1× A100-SXM4-40GB	WatsonX
Judge LLM	Llama-4-Maverick-17B (WatsonX) · 6-criterion rubric · threshold 0.6	
Trials per cell	6 (matched final-six A100 cohort)	
Corpus	61 canonical scenarios + 5 negative validation fixtures	
Evidence floor	31-scenario post-PR175 judged set unless a caption says otherwise	
Tracking	W&B project assetopsbench-smartgrid	
Software stack	vLLM · PyTorch · MCP (stdio) · nvidia-smi · torch.profiler	

70B note. Hosted Llama-3.3-70B (V-M-70) matched 8B judge mean but ran at p95 ≈ 271s vs ≈ 58s for the 8B core. We keep it as scale-sanity, never folded into the 8B matrix without explicit per-cell n labels.